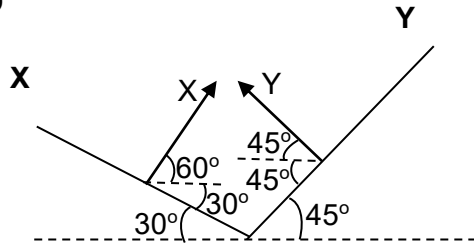


By velocity of approach and separation, KE not conserved for option D

6

D



consider horizontal equilibrium,

$$X_x = Y_x$$

$$X \cos 60^\circ = Y \cos 45^\circ$$

$$\frac{X}{Y} = 1.41$$

7

A

Since object is thrown at high speed, it experiences large upward viscous force, so